

Daniel Amirifard

AI/ML Engineer

Contact

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Professional Summary

An **AI/ML Engineer** with **4+** years of experience in **Generative AI**, **Large Language Models**, and **ML/DL** domains. Graduated from **Politecnico di Milano** and **delivered 30+** AI projects across academia and industry. Experienced in building **scalable** AI systems, LLM applications, RAG pipelines, **agentic workflows**, and **production-ready model** deployment, with collaborations and experience connected to **STMicroelectronics**, **Nokia**, and **Baker Hughes** companies.

Skills

Programming: Python, SQL

Machine Learning & AI: TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib

Generative AI & LLMs: RAG, Agentic AI, LangChain Prompt Engineering, Chatbots

Cloud & DevOps: AWS (Amazon Bedrock, S3), Azure ML Studio, Linux, Git, GitHub, Docker, FastAPI, CI/CD

Others: English (C1), Italian (A2), Teamwork, Organization, Communication, Creativity

Work Experience

Artificial Intelligence Engineer

11/2025 – Present

🔗 **Hermes/Baker Hughes**, Full-time

Milan, Italy

- Developed **3+ production-oriented GenAI solutions** from PoC to deployment, using **RAG pipelines**, **LangGraph**, and **Amazon Bedrock** to improve automation, scalability, and business impact.
- Built **agentic AI systems** with **multi-step reasoning**, **workflow automation**, **prompt engineering**, and **decision-making pipelines**, reducing manual analysis around **85%** across client-facing business processes.
- Delivered AI applications across **3 industry domains**, including **financial invoice classification**, **chemical-sector workflows**, and **energy-sector applications**, with one ML classification pipeline achieving a **93% F1-score**.

Machine Learning Engineer

05/2025 – 11/2025

🔗 **STMicroelectronics**, Full-time

Milan, Italy

- Designed and deployed automated **neural architecture search (NAS)** workflows for **LLM agent bots**, improving model accuracy and computational efficiency by 40%.
- Implemented **zero-cost proxy** optimization methods, accelerating **generative AI bench-marketing** by 10X, while lowering resource consumption.
- Constructed and integrated an **AutoNN LLM agent**, that automatically designs neural network structures, reducing model architecture generation time 70%.

Machine Learning Engineer

02/2023 – 10/2024

🔗 **Polimi Data Scientists Association**, Part-time

Milan, Italy

- Developed, trained, and optimized 10+** machine learning models in Computer vision, anomaly detection and time series.
- Organised data **pipelines** for efficient **ETL** processes and source relevant datasets, adopted in 5+ projects like SIIT and Breast Cancer detection.
- Showcased project progress to team members and international students biweekly, successfully leading to the completion of **4 significant projects**, using **GenAI**.

Data Scientist

09/2021 - 09/2022

Doran Smart-City, Full-time

Tehran, Iran

- Executed machine learning algorithms like **Random Forest** and **XGBoost** to analyze and model urban traffic data, resulting in a **15% improvement in traffic flow**.
- Redesigned predictive models using gradient boosting and random forests for smart energy management, leading to a **10% reduction in energy consumption** within city infrastructure.

Education

Polytechnic University of Milan M.Sc. Telecommunication Engineering – Computer Science and communications	09/2022 - 03/2026 Milan, Italy
Amirkabir University of Technology B.Sc. Engineering	09/2016 - 09/2021 Tehran, Iran

Projects

- RAG-Based Financial Invoice Classification Pipeline** | Amazon Bedrock, GenAI, Agentic-systems, LangGraph
- Built an AI-powered invoice classification pipeline using RAG, FAISS, LangGraph, and Mistral via Amazon Bedrock, automating milestone classification from invoice text fields and achieving a 93% F1-score.
-  **Long Form Question Answering ChatBot** | Retrieval-augmented generation (RAG), LLAMA, LoRA, BART, NLP, Hugging Face
- Created an NLP pipeline for Long Form Question Answering using transformers (BART, LLAMA) and RAG in a team of 4, securing a ROUGE score of 22.
- Zero-Cost Proxies in AutoNN LLM Agent** | Google Cloud, Azure ML Studio, GenAI, Fine Tuning, Agentic AI Systems, OpenAI
- Integrated zero-cost proxies into AutoNN LLM agent, achieving 0.78 Kendall’s Tau on CIFAR-x datasets for reliable training-free architecture performance estimation.
- SIIT ChatBot** | RAG, Chain-of-Thought, GPT, BERT, LLAMA, LangChain, Pytorch, FAISS, Prompt Engineering
- Spearheaded a technical team of 10 in the SIIT project with JEMP startup for SIIT Company, revolutionizing pharmaceutical synthesis and slashing production time by 25% by developing a chatbot.
-  **Cooperative Perception In Automated Cars** | 3D Object detection, GenAI, Transformers
- Developed a V2V cooperative perception pipeline using OPV2V, LiDAR, PointPillars, ROI filtering, and attention fusion, achieving 0.914 AP@0.5 and 0.837 AP@0.7 for 3D object detection.
-  **Traffic Forecasting** | Time Series, Data Processing, Decision trees, RNN, LSTM, Conv Nets, Mlflow
- Directed a team of 3 to achieve a 9% MAPE in estimating 50 Mbits/s channels using GEANT Dataset with RNN and CONV–LSTM networks.
- Other Projects** | R-CNN, Yolo, Mask learning, GANs,VAEs, GenAI, MLflow
-  ClimateFish Clustering,  Mac-Derandomization,  Application Identifier,  Breast Cancer Detection,  ML Intrusion Detection,  Univariate Time Series Forecasting,  Plant Classification (Healthy vs Unhealthy)

Workshop Experience

Impact of non-Gaussian noise on LDPC performance Project Nokia Company	06/2024 Vimercate, Italy
Presented to over 100 attendees at Nokia, showcasing a simulation of non-Gaussian noise effects on LDPC performance using QAM 4, 16, and 64.	
MultiNet-Route-Optimizer Project Nokia Company	03/2024 Vimercate, Italy
Presented a 65% improvement in routing efficiency for a Nokia-defined problem, achieved by a 4-person team using Top-Down and Bottom-Up optimization strategies.	

Certifications

- Neural Networks and Deep Learning** | DeepLearning.AI
- Supervised Learning** | From DeepLearning.AI
- Linux LPIC 1** | Tehran Institute of Technology
- Python** | Tehran Institute of Technology